

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited           |
| <b>Originating Company</b>     | Epic Atlantic Limited                                        |
| <b>Document Title</b>          | Wake Frequency Calculation- BBOU/ELEP/AKAB                   |
| <b>Document Number</b>         | EA-BFPCL-GPMC-GGPL-FEED-ICS-RPT-002                          |
| <b>Document Revision</b>       | R02                                                          |
| <b>Document Status</b>         | Issued For Review                                            |
| <b>Originator</b>              | P.Kiran                                                      |
| <b>Security Classification</b> | Restricted                                                   |
| <b>Issue Date</b>              | 26-Oct-22                                                    |

Wake Frequency Calculation- BBOU/ELEP/AKAB

|              |                   |                           |                   |                 |                        |                         |
|--------------|-------------------|---------------------------|-------------------|-----------------|------------------------|-------------------------|
|              |                   |                           |                   |                 |                        |                         |
|              |                   |                           |                   |                 |                        |                         |
|              |                   |                           |                   |                 |                        |                         |
| R02          | 26.10.22          | Re-Issued for Review      | P.Kiran           | P.Roy           | W.Irem                 |                         |
| R01          | 28.06.22          | Issued for Review         | P.Kiran           | P.Roy           | W.Irem                 |                         |
| <b>Rev #</b> | <b>Issue Date</b> | <b>Status Description</b> | <b>Originator</b> | <b>Reviewer</b> | <b>Approver (Epic)</b> | <b>Approver (BFPCL)</b> |

**Revision History**

| Rev No. | Date       | Description of Revision                               |
|---------|------------|-------------------------------------------------------|
| R01     | 6/28/2022  | Issued for Review                                     |
| R02     | 10/26/2022 | Re-Issued for Review after incorporating TCE comments |
|         |            |                                                       |

**Revision Philosophy:**

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

**Restricted**

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL)  
© All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

## SUMMARY PAGE

|                  |                                                     |                                                       |
|------------------|-----------------------------------------------------|-------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                       |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                             |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED-ICS-RPT-002 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

**KEYWORDS:** BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

## REFERENCES

| Document No. | Description |
|--------------|-------------|
|              |             |
|              |             |
|              |             |
|              |             |
|              |             |
|              |             |
|              |             |
|              |             |
|              |             |

---

**Table of Contents**

|                        |   |
|------------------------|---|
| Wake Frequency Sheet 1 | 5 |
| Wake Frequency Sheet 2 | 6 |

| Sr. No. | Tag No.          | Application                                             | Pipe OD Inch | Pipe OD in mm | Pipe schedule | Thickness in mm | Process fluid | fluid Density (g/cm3) | Immeson Length (mm) | Ovalall Length TW length (mm) with 150 nozzle height | Wake Frequency Pass or fail |
|---------|------------------|---------------------------------------------------------|--------------|---------------|---------------|-----------------|---------------|-----------------------|---------------------|------------------------------------------------------|-----------------------------|
| 1       | BBOU-TW/TIT-1001 | Battery Limit Temperature at the Inlet Manifold of BBOU | 24"          | 610           | 100           | 26.2            | Gas           | 0.811                 | 125                 | 275                                                  | Pass                        |
| 2       | BBOU-TW/TI-1001  | Battery Limit Temperature at the Inlet Manifold of BBOU | 24"          | 610           | 100           | 26.2            | Gas           | 0.811                 | 125                 | 275                                                  | Pass                        |
| 3       | ELEP-TW/TIT-1001 | Battery Limit Temperature at the Inlet Manifold of ELEP | 36"          | 914           | -             | 38.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |
| 4       | ELEP-TW/TI-1001  | Battery Limit Temperature at the Inlet Manifold of ELEP | 36"          | 914           | -             | 38.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |
| 5       | ELEP-TW/TIT-1002 | Battery Limit Temperature-Pig Receiver ELEP-A-200       | 24"          | 610           | 100           | 26.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |
| 6       | ELEP-TW/TI-1002  | Battery Limit Temperature-Pig Receiver ELEP-A-200       | 24"          | 610           | 100           | 26.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |
| 7       | AKAB-TW/TIT-1001 | Battery Limit Temperature-Pig Receiver AKAB-A-100       | 36"          | 914           | -             | 38.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |
| 8       | AKAB-TW/TI-1001  | Battery Limit Temperature-Pig Receiver AKAB-A-100       | 36"          | 914           | -             | 38.2            | Gas- Liquid   | 0.811                 | 125                 | 275                                                  | Pass                        |

## Notes :

1. Nozzle Standout = 15 0 mm
2. Nozzle Size = 1.5", 900#
3. Fluid Velocity = 5 m/s
4. Thermowell Material = SS316
5. Operating Pressure = 100 kg/cm<sup>2</sup>g
6. Operating Temperature=100
7. Pipe Class= 900#, C03
8. Fluid Density = 811 kg/m<sup>3</sup>
9. Tip Diameter= 18 mm
10. Tip Thickness= 6mm
11. If the velocity is higher than 5 m/s, it is recommended to have helical vortex shedding thermowell.
12. According to API RP 551, the immersion length of thermowells in the pipes ( the length from well tip to pipe wall) shall be maximum 5" or 125mm.
13. The immersion length in vessels liquid shall be 400 mm.
14. In vessels containing gas, the immersion length shall be according to :  
Vessel Dia (D) upto 800 mm : TW length: 200 mm, Vessel Dia (D) more than 800 mm less than equal 100mm, TW length: 300 mm
15. The insertion length depends on the length of the nozzle or pipe fitting, taking any possible refractory lining into account.

|  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |

## WAKE FREQUENCY ANALYSIS

PER ASME PTC 19.3 TW-2016

**PASS**

### THERMOWELL

Unit Type: **Metric** ▼Code: **B31.3** ▼  
(for allowable stresses)Type: **Flanged Full Pen Weld** ▼Stem Style: **Tapered** ▼(U) Insertion: **275** mm(Q) Root Diameter: **22** mm(V) Tip diameter: **18** mm(B) Bore Diameter: **7** mm(TT) Tip Thickness: **8** mm(SL) Shield Length: **150** mmFlange Class: **900** ▼Flange Material: **316 SS** ▼TW Material: **316 SS** ▼(b) Root Fillet Radius: **3.175** mm  
(.125" is standard)OAL Length: **355** mm

### FLUID

(DN) Density: **811** kg/m<sup>3</sup>.(VISC) Viscosity: **0.03** cP(T) Temperature: **100** C(P) Pressure: **100** bar(VEL) Velocity: **5** m/s**Calculate**